

Direct, Convergent α -Amidation of β -Keto Esters via Nickelocene Catalysis

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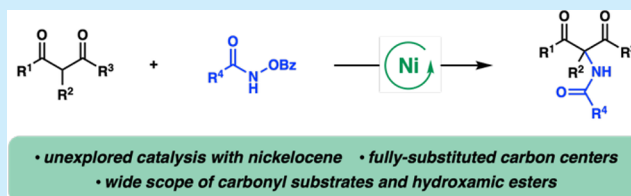


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Supporting Information

ABSTRACT: α -Amidation of carbonyl-containing compounds without prefunctionalization of starting materials represents a continuing challenge within the scientific community. Here, nickelocene is used catalytically to effect an amidation of β -keto esters. Broad substrate tolerance on both coupling partners is achieved, enabling the application toward a convergent synthetic strategy. Initial mechanistic investigations suggest complex activation pathways.



Nitrogen functionality α to carbonyl groups, beyond the 20 naturally occurring amino acids, is found in many natural products and pharmacologically active materials (Figure 1A).^{1,2} To enable a desirable synthetic strategy for access to this class of compounds, introduction of an amine or amide functionality at a late stage may be appealing. However, most methods to do so currently rely on either prefunctionalization of the carbonyl moiety³ or require introducing nitrogen in a higher oxidation state,⁴ subsequently necessitating reduction and lengthening a synthetic sequence. Avoiding this additional manipulation through the development of methods to directly amidate unfunctionalized carbonyl-containing compounds is thus of importance to the synthetic community.

Advances have been made in the area of α -amination, with examples utilizing both transition metal catalysis and biocatalysis, although these strategies struggle to accommodate electron-withdrawing substituents.⁵ Recent advances overcoming this challenge have enabled the incorporation of carbamates and *N*-tosylates by leveraging active intermediates, such as nitrenes,^{6,7} *N*-centered radicals,⁸ or reaction with α -oxidized carbonyl compounds.⁹ While these methods all represent key advances in the strategy for the direct introduction of nitrogen α to carbonyls, further sequential transformations would be necessary to introduce complexity on the nitrogen moiety. In an ideal convergent coupling strategy, a functionalized amide could be brought in directly, forging this key bond in a more advantageous fashion for synthetic efficiency (Figure 1B).

To this end, methodology has been disclosed by the groups of Yu and Bao,¹⁰ Zong, Jiang, and Qiu,¹¹ and Chang¹² using iron catalysis, which in conjunction with photochemistry can achieve α -amidation (Figure 1C). This reaction manifold, however, suffers from challenges, largely revolving around sterics at the α position; either reactivity is limited when α substitution is included, or α substitution is needed to prevent

double amidation. Additionally, a broad substrate tolerance on both coupling partners was a challenge in all reports. As a result, an alternate catalytic system allowing for selective yet reactive amidation is thus desirable. Here, we demonstrate a system capable of achieving selective amidation of β -keto esters with broad functional group tolerance on both coupling partners (Figure 1D).

We began our investigations by turning to a previous report from our group in which we leveraged a proposed nitrene intermediate as an electrophilic amide source for the synthesis of hydrazides.¹³ We hypothesized that trapping this intermediate with an enolate would forge the desired bond under nickel catalysis. In an effort to develop a reaction capable of forming fully substituted carbon centers, the reactivity between ethyl 2-methyl-3-oxobutanoate (**1a**) and *N*-(benzoyloxy)-benzamide (**2a**) was explored. To our delight, the initial investigation using nickel half-sandwich catalyst **4a** and lithium *tert*-butoxide as a base resulted in the formation of the desired amidated product **3a**, albeit in a low yield (Table 1, entry 1). Reinvestigation of half-sandwich catalysts revealed that compound **4b**, with bulkier NHC substitution, improved the yield to 36% (entry 2). The solvent mixture of dichloromethane and tetrahydrofuran (THF), while vital for achieving high yields in previous nickel chemistry,¹³ was no longer optimal in this system, and 1,4-dioxane was found to both improve the yield and reaction time (entry 4). Alternate bases were investigated (entries 5 and 6), although lithium *tert*-butoxide remained optimal. Interestingly, a counterion effect

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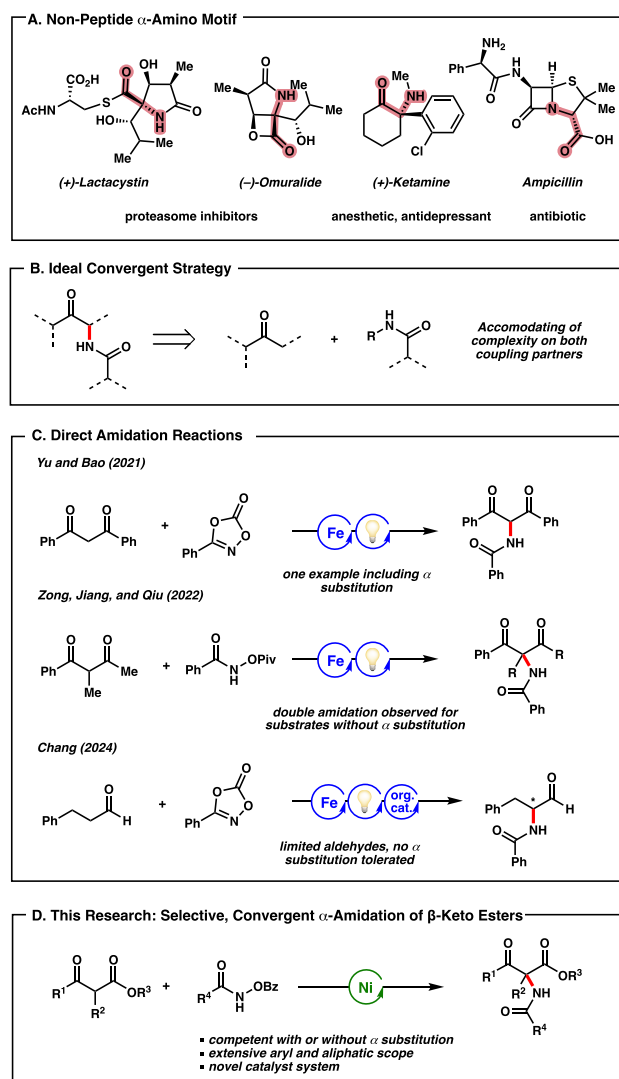


Figure 1. (A) Natural products and pharmaceuticals displaying the α -amino or amido motif. (B) Idealized umpolung amidation transformation. (C) Current strategies for the direct amidation of carbonyl compounds. (D) Our strategy toward α -amidation.

was observed for the base, with potassium *tert*-butoxide failing to generate more than a small amount of product. Alternate silane reductants were explored (entries 7 and 8), although phenylsilane proved superior. Further investigation of additives, including equivalents of silane,¹⁵ can be found in the [Supporting Information](#).

At this stage, a few general reaction mechanisms were explored. To examine if this reaction proceeds through an uncatalyzed S_N2 reaction at nitrogen, preceded for alternate N–O species,¹⁶ the catalyst was omitted (entry 9). As no product was observed in the absence of a nickel catalyst, this pathway was rejected. The possibility of nickel acting as a Lewis acid to activate the starting materials was also explored; replacing the nickel catalyst with catalytic amounts of a variety of Lewis acids resulted in no product formation (see the [Supporting Information](#)), leading to nickel catalysis being favored as the most likely manifold.

Further mechanistic analysis revealed key differences between this transformation and our previous report on the nickel-catalyzed synthesis of hydrazides.¹³ Previously, a Ni(I) active catalyst was proposed, with the role of phenylsilane

Table 1. Optimization of α -Amidation Using Nickel Half-Sandwich Complexes

entry	[Ni]	base	additive	solvent	yield (%)
1	[Ni] 4a	LiOt-Bu	PhSiH ₃	1:4 CH ₂ Cl ₂ /THF	7 ^a
2	[Ni] 4b	LiOt-Bu	PhSiH ₃	1:4 CH ₂ Cl ₂ /THF	36 ^a
3	[Ni] 4b	LiOt-Bu	PhSiH ₃	THF	33 ^a
4	[Ni] 4b	LiOt-Bu	PhSiH ₃	1,4-dioxane	74
5	[Ni] 4b	KOt-Bu	PhSiH ₃	1,4-dioxane	5
6	[Ni] 4b	TBD	PhSiH ₃	1,4-dioxane	29
7	[Ni] 4b	LiOt-Bu	Ph ₂ SiH ₂	1,4-dioxane	23
8	[Ni] 4b	LiOt-Bu	PMHS	1,4-dioxane	—
9	—	LiOt-Bu	PhSiH ₃	1,4-dioxane	—
10	[Ni] 4c	LiOt-Bu	—	1,4-dioxane	—

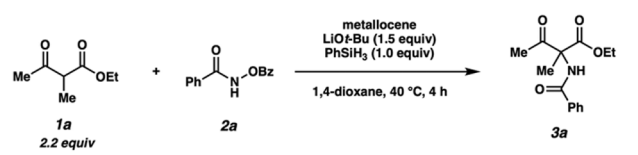
4a 4b 4c 5

^aRequiring 24 h.

hypothesized to be reduction of the Ni(II) precatalyst.¹⁷ In the current study, however, upon preparing Ni(I) catalyst 4c and subjecting it to reaction conditions in the absence of phenylsilane, no reaction was observed (entry 10). Additionally, throughout the optimization of this amidation reaction, side products analogous to compound 5 were observed,¹⁸ presumed to arise through a deleterious side reaction between the starting material and catalyst. Despite this use of the same hydroxamic ester and catalyst in our previous report,¹³ these side products were never observed, further suggesting mechanistic divergence.

Following several unsuccessful attempts to induce enantioselectivity through the inclusion of asymmetric ligands, we began to consider the possibility of off-metal redox activation. We were drawn to previous reports suggesting that ferrocene is capable of N–O bond cleavage through single-electron transfer from ferrocene, generating imidyl¹⁹ and amidyl²⁰ radicals, as well as a precedent for amidyl radicals as intermediates in α amidation.^{8,10} As metallocene catalysts were an appealing alternative to avoid the observed NHC–hydroxamate adduct (5), nickelocene was investigated for its competence in the reaction. Gratifyingly, nickelocene led to similar yields ([Table 2](#), entry 1), with lower catalyst loadings now preferred (entry 2). Notably, in comparison to half-sandwich catalysis, markedly shorter reaction times were observed for other β -keto ester substrates ([Supporting Information](#)). Decreasing the base loading slightly and increasing the temperature led to a now optimal 82% yield (entry 3); performing this reaction on a 1.0 mmol scale led to a similar yield of 78% (entry 4), demonstrating good scalability. Furthermore, while ferrocene and cobaltocene both led to product formation, they were slower and less effective than nickelocene (entries 5 and 6). This represents a rare example of nickelocene in catalysis, despite its longstanding study in inorganic chemistry literature.

The scope of the reaction was then investigated ([Scheme 1](#)). The α substituent could be replaced with an ethyl with no decrease in the yield observed (3b). β -Keto esters with no α -substitution also performed well (3c and 3d), with good yields

Table 2. Optimization of α -Amidation Using Metallocene Catalysts


entry	metallocene	equivalents	yield (%)
1	NiCp ₂	10 mol%	65 ^a
2	NiCp ₂	5 mol%	69 ^a
3	NiCp ₂	5 mol%	82
4	NiCp ₂	5 mol%	78 ^b
5	FeCp ₂	5 mol%	70 ^c
6	CoCp ₂	5 mol%	56 ^d

^a2.0 equiv LiOt-Bu, 30 °C. ^bOn a 1.0 mmol scale. ^cRequiring 24 h.^dRequiring 48 h.

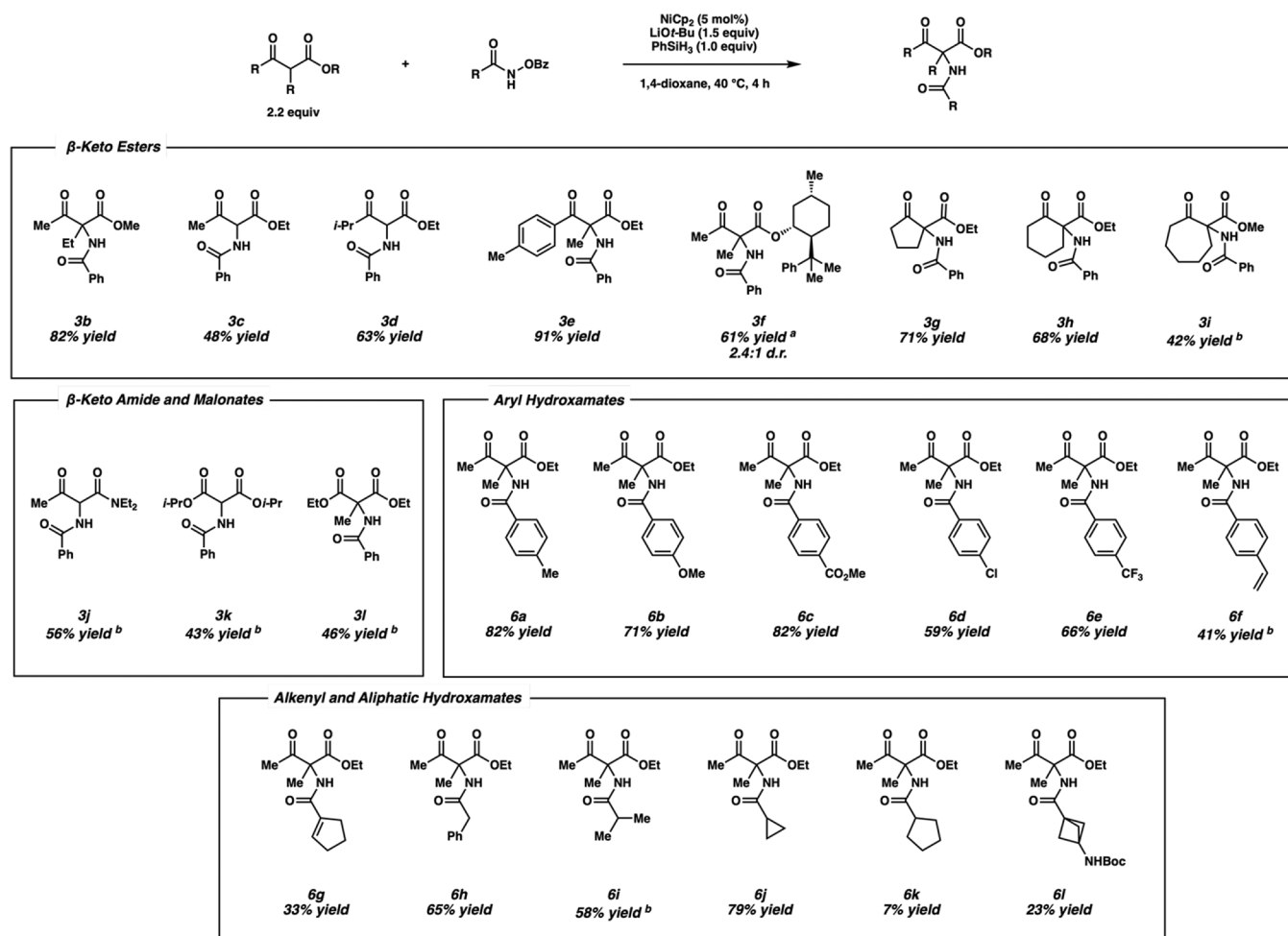
of the singly amidated product. Aryl β -keto esters also formed the product in a high yield (3e). Inclusion of a chiral auxiliary resulted in product formation with some diastereocontrol (3f). Cyclic β -keto esters also performed well, with 5- through 7-membered rings resulting in good yields (3g–3i). While both were electronically differentiated from the standard material, β -

keto amides (3j) and malonates (3k and 3l) were competent in the reaction.

On the hydroxamate starting material, electron-donating and electron-withdrawing substituents could be added with little impact on the yield (6a–6c). Substitution including halogens, either *p*-Cl (6d) or *p*-CF₃ (6e), did lead to a decrease, though good yields were still produced. Styrenyl substitution could additionally be included (6f), with no competing side reactivity observed. Alkenyl and aliphatic substituents were also competent in the reaction (6g–6l). Notably, benzyl substitution (6h) is well-tolerated in this reaction, contrasting with our group's previous report where the starting material was inert to reaction conditions¹³ as well as further indicating selectivity against benzylic functionalization. Cyclic alkyl substituents displayed a dependence on ring sizes, with smaller cyclopropyl hydroxamates outperforming cyclopentyl (6j and 6k) (*vide infra*).

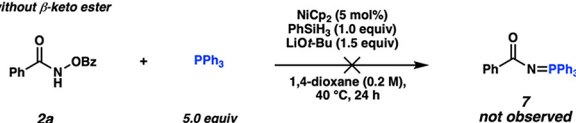
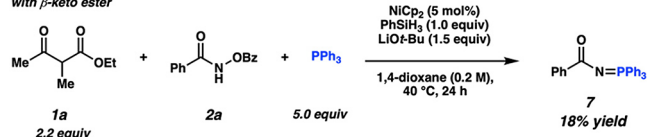
To further investigate the mechanism of this reaction, we explored two main reaction manifolds proposed for prior iron-catalyzed transformations, either a nitrene pathway^{11,12} or a radical pathway.¹⁰ Beginning by exposing the reaction conditions to triphenylphosphine to intercept a potential nitrene intermediate (Scheme 2A), different outcomes were observed depending on the inclusion or exclusion of β -keto ester 1a. In the absence of compound 1a, no iminophosphor-

Scheme 1. Substrate Scope of the Amidation Reaction

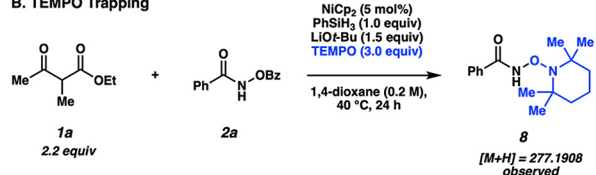
^aRequiring 5 h. ^bRequiring 24 h.

Scheme 2. Trapping Experiments

A. Triphenylphosphine Trapping

without β -keto esterwith β -keto ester

B. TEMPO Trapping



ane 7 was observed, although upon inclusion of β -keto ester **1a**, compound **7** was formed in 18% yield, alongside a small amount of amidated product **3a**. This contrasting activity may suggest a role for the β -keto ester in the formation of a reactive nitrogen species. Additionally, while the formation of an iminophosphorane is traditionally associated with trapping of an active nitrene intermediate,²¹ a recently proposed mechanism for its formation involves interception of a N-centered radical with phosphorus, followed by oxidation,²² suggesting that the formation of this side product is inconclusive.

In an attempt to trap a key radical intermediate, TEMPO was included in the reaction conditions (Scheme 2B). While the resulting species was too unstable for isolation, the mass of the potential TEMPO adduct formed with a N-centered radical originating from hydroxamate **2a** was observed via high-resolution mass spectrometry (HRMS) analysis. These experiments taken alongside precedent analogues reactivity with ferrocene led us to hypothesize the intermediacy of a N-centered radical species. Such an intermediate would also suggest an explanation for the different outcomes observed for the formation of compounds **6i** and **6j** vs compound **6k**, as the cyclopentyl substrate is large enough to potentially undergo a deleterious 1,5-hydrogen atom transfer inaccessible to either the isopropyl or cyclopropyl starting materials.

In conclusion, we have developed a nickel-catalyzed amidation of β -keto esters, enabling the convenient and convergent synthesis of α -amidated materials. Both aliphatic and aromatic substituents are well-tolerated on either coupling partner. The use of nickelocene in catalysis is disclosed, with favorable reactivity in comparison to other metallocenes. A N-centered radical is hypothesized as a reactive intermediate from the initial mechanistic investigation. Further experimental and computational investigations into the mechanism are ongoing.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.orglett.5c01621>.

Experimental procedures and spectroscopic data (PDF)

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Notes

The authors declare no competing financial interest.

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